

2.1

Bisection Method

* IVT: if $f(a)$ and $f(b)$ are opposites in sign, then there is a $c \in [a, b]$, such that $f(c) = 0$ $f(a) = +ve \rightarrow f(b) = -ve$

* Finding the approximate value of x for which $f(x) = y = 0$

* Finding approximate solution of a function * We try to decrease the interval by dividing it by 2 each time till we find c

Bisection Technique The first technique, based on the Intermediate Value Theorem, is called the Bisection, or Binary-search, method. Suppose f is a continuous function defined on the interval $[a, b]$, with $f(a)$ and $f(b)$ of opposite sign.

The Intermediate Value Theorem implies that a number p exists in (a, b) with $f(p) = 0$.

Although the procedure will work when there is more than one root in the interval (a, b) , we assume for simplicity that the root in this interval is unique. The method calls for a repeated halving (or bisecting) of subintervals of $[a, b]$ and, at each step, locating the half containing p . To begin, set $a_1 = a$ and $b_1 = b$ and let p_1 be the midpoint of $[a, b]$, that is.

$$p_1 = a_1 + \frac{b_1 - a_1}{2}$$

If $f(p_1) = 0$, then $p = p_1$, and we are done.

If $f(p_1) \neq 0$, then $f(p_1)$ has the same sign as either $f(a_1)$ or $f(b_1)$.

o If $f(p_1)$ and $f(a_1)$ have the same sign, $p \in (p_1, b_1)$. Set $a_2 = p_1$ and $b_2 = b_1$.

o If $f(p_1)$ and $f(b_1)$ have opposite signs, $p \in (a_1, p_1)$. Set $a_2 = a_1$ and $b_2 = p_1$. Then reapply the process to the interval $[a_2, b_2]$

Bisectional method steps:

① check $f(a), f(b)$ [opposite in sign]

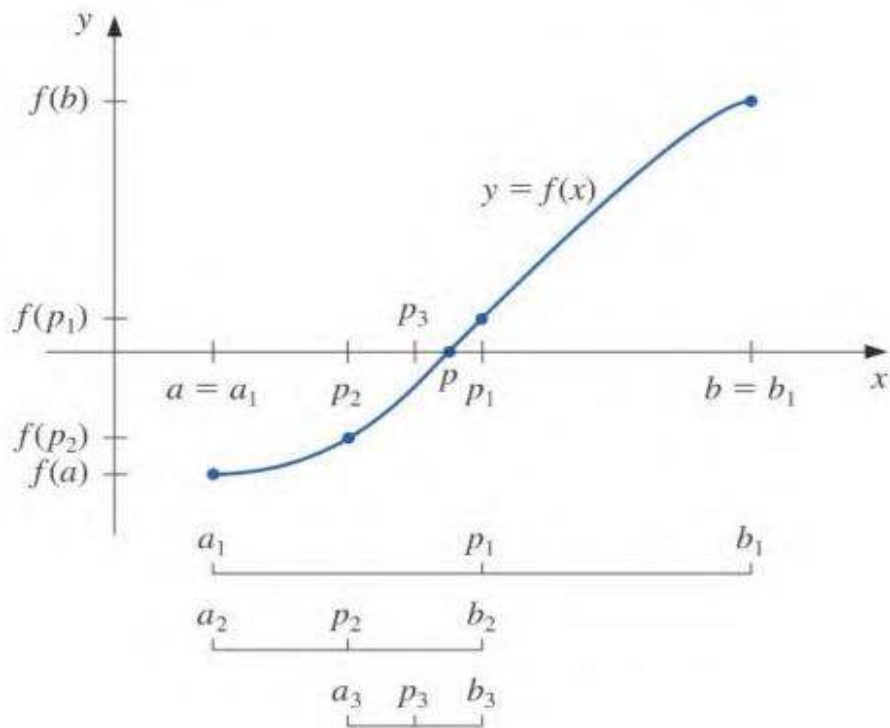
② $P_1 = \frac{a+b}{2}$

③ $f(P_1) \rightarrow$ check the sign

* Sign of $f(P_1)$ is the same sign of $f(a)$ then replace a by P_1 $P_2 = \frac{P_1+b}{2}$ new interval $[P_1, b]$

* else if the sign of $f(P_1)$ is the same as sign of $f(b)$ then replace b by P_1 $P_2 = \frac{a+P_1}{2}$, new interval $[a, P_1]$

④ $f(P_2)$ check the sign and do the same as step 3 till the P number given to stop



EXERCISE SET 2.1

1. Use the Bisection method to find p_3 for $f(x) = \sqrt{x} - \cos x = 0$ on $[0, 1]$.
2. Let $f(x) = 3(x+1)(x - \frac{1}{2})(x-1) = 0$. Use the Bisection method on the following intervals to find p_3 .
 - a. $[-2, 1.5]$
 - b. $[-1.25, 2.5]$
3. Use the Bisection method to find solutions accurate to within 10^{-2} for $x^3 - 7x^2 + 14x - 6 = 0$ on each interval.
 - a. $[0, 1]$
 - b. $[1, 3.2]$
 - c. $[3.2, 4]$
4. Use the Bisection method to find solutions accurate to within 10^{-2} for $x^4 - 2x^3 - 4x^2 + 4x + 4 = 0$ on each interval.
 - a. $[-2, -1]$
 - b. $[0, 2]$
 - c. $[2, 3]$
 - d. $[-1, 0]$
5. Use the Bisection method to find solutions accurate to within 10^{-5} for the following problems.
 - a. $x - 2^{-x} = 0$ for $0 \leq x \leq 1$
 - b. $e^x - x^2 + 3x - 2 = 0$ for $0 \leq x \leq 1$
 - c. $2x \cos(2x) - (x+1)^2 = 0$ for $-3 \leq x \leq -2$ and $-1 \leq x \leq 0$
 - d. $x \cos x - 2x^2 + 3x - 1 = 0$ for $0.2 \leq x \leq 0.3$ and $1.2 \leq x \leq 1.3$
6. Use the Bisection method to find solutions, accurate to within 10^{-5} for the following problems.
 - a. $3x - e^x = 0$ for $1 \leq x \leq 2$
 - b. $2x + 3 \cos x - e^x = 0$ for $0 \leq x \leq 1$
 - c. $x^2 - 4x + 4 - \ln x = 0$ for $1 \leq x \leq 2$ and $2 \leq x \leq 4$
 - d. $x + 1 - 2 \sin \pi x = 0$ for $0 \leq x \leq 0.5$ and $0.5 \leq x \leq 1$
7.
 - a. Sketch the graphs of $y = x$ and $y = 2 \sin x$.
 - b. Use the Bisection method to find an approximation to within 10^{-5} to the first positive value of x with $x = 2 \sin x$.
8.
 - a. Sketch the graphs of $y = x$ and $y = \tan x$.
 - b. Use the Bisection method to find an approximation to within 10^{-5} to the first positive value of x with $x = \tan x$.
9.
 - a. Sketch the graphs of $y = e^x - 2$ and $y = \cos(e^x - 2)$.

$$\textcircled{1} f(x) = \sqrt{x} - \cos x = 0 \quad [0, 1]$$

$$f(a) = f(0) = \sqrt{0} - \cos 0 = -ve$$

$$f(b) = f(1) = \sqrt{1} - \cos 1 = +ve$$

By IVT, there is a $c \in [0, 1]$ such that $f(c) = 0$

$$P_1 = \frac{0+1}{2} = \frac{1}{2} = 0.5 \text{ (same sign as } b)$$

$$f(0.5) = \sqrt{0.5} - \cos(0.5) = -ve$$

Replace a by P_1 $[0.5, 1]$

$$P_2 = \frac{0.5+1}{2} = \frac{3}{4} = 0.75 \text{ (same sign as } b)$$

$$f(0.75) = \sqrt{0.75} - \cos(0.75) = -ve \quad [0.5, 0.75]$$

$$P_3 = \frac{0.5+0.75}{2} = \frac{5}{8} = 0.625$$

If we need to find P_4

$$f(0.625) = \sqrt{0.625} - \cos 0.625 = -ve \quad [0.625, 0.75]$$

$$P_4 = \frac{0.625+0.75}{2} = \frac{11}{16} = 0.6875$$

$$\textcircled{3} f(x) = x^3 - 7x^2 + 14x - 6 = 0$$

$$\textcircled{C} [3.2, 4]$$

$$f(3.2) = (3.2)^3 - 7(3.2)^2 + 14(3.2) - 6 = -ve$$

$$f(4) = (4)^3 - 7(4)^2 + 14(4) - 6 = +ve$$

By IVT, there is a $c \in [3.2, 4]$ such that $f(c) = 0$

$$P_1 = \frac{3.2+4}{2} = 3.6$$

$$f(3.6) = (3.6)^3 - 7(3.6)^2 + 14(3.6) - 6 = +ve \quad [3.2, 3.6]$$

$$P_2 = \frac{3.2+3.6}{2} = 3.4$$

$$f(3.4) = (3.4)^3 - 7(3.4)^2 + 14(3.4) - 6 = +ve \quad [3.4, 3.6]$$

$$P_3 = \frac{3.2+3.6}{2} = 3.5$$

$$f(3.5) = (3.5)^3 - 7(3.5)^2 + 14(3.5) - 6 = +ve \quad [3.4, 3.5]$$

$$P_4 = \frac{3.4+3.5}{2} = 3.45$$

$$|P_4 - P_3| = 0.05$$

$$f(3.45) = +ve \quad [3.4, 3.45]$$

$$|P_5 - P_4| = 0.025$$

$$P_5 = \frac{3.4+3.45}{2} = 3.425$$

$$f(3.425) = +ve \quad [3.4, 3.425]$$

$$|P_6 - P_5| = 0.0125$$

$$P_6 = \frac{3.425+3.4}{2} = 3.4125$$

$$f(3.4125) = -ve \quad [3.4125, 3.425]$$

$$|P_7 - P_6| = 0.00625 \leq 10^{-2}$$

$$P_7 = \frac{3.4125+3.425}{2} = 3.41875$$

P_7 is the approximate solution

$$P_1, P_2, P_3, P_4$$

$$(P_3 - P_2) \leq 10^{-2}$$

$$(P_4 - P_3) \leq$$

5. Use the Bisection method to find solutions accurate to within 10^{-5} for the following problems.

a. $x - 2^{-x} = 0$ for $0 \leq x \leq 1$ (upto 10^{-3})

$$f(0) = 0 - 2^{-0} = -1 \text{ (-ve)}$$

$$f(1) = 1 - 2^{-1} = \frac{1}{2} \text{ (+ve)}$$

$[-ve, +ve]$

$$P_1 = \frac{0+1}{2} = \frac{1}{2} = 0.5$$

$$f\left(\frac{1}{2}\right) = \frac{1}{2} - 2^{-\frac{1}{2}} = -0.2 \text{ (-ve)} [0.5, 1]$$

$$P_2 = \frac{\frac{1}{2}+1}{2} = \frac{3}{4} = 0.75$$

$$f\left(\frac{3}{4}\right) = \frac{3}{4} - 2^{-\frac{3}{4}} = 0.155 \text{ (+ve)} [0.5, 0.75]$$

$$P_3 = \frac{\frac{1}{2} + \frac{3}{4}}{2} = \frac{5}{8} = 0.625$$

b. $e^x - x^2 + 3x - 2 = 0$ for $0 \leq x \leq 1$

upto 7th iteration ($P_7 = 0.2578125$)
upto 10^{-2}

$$f(0) = -1 \text{ (-ve)} \quad [-ve, +ve]$$

$$f(1) = 2.7 \text{ (+ve)}$$

$$P_1 = \frac{0+1}{2} = 0.5$$

$$f(0.5) = 0.9 \text{ (+ve)} [0, 0.5]$$

$$P_2 = \frac{0+0.5}{2} = 0.25$$

$$f(0.25) = -0.03 \text{ (-ve)} [0.25, 0.5]$$

$$P_3 = \frac{0.25+0.5}{2} = 0.375$$

$$f(0.375) = 0.44 \text{ (+ve)} [0.25, 0.375]$$

$$P_4 = \frac{0.25+0.375}{2} = 0.3125$$

$$f(0.3125) = 0.2 \text{ (+ve)} [0.25, 0.3125]$$

$$P_5 = \frac{0.25+0.3125}{2} = 0.28125$$

$$f(0.28125) = 0.32 \text{ (+ve)} [0.25, 0.28125]$$

$$P_6 = \frac{0.25+0.28125}{2} = 0.265625$$

$$f(0.265625) = +ve [0.25, 0.265625]$$

$$P_7 = \frac{0.25+0.265625}{2} = 0.2578125$$

★ The process we used
is called iteration

$$|P_4 - P_3| = |0.3125 - 0.375| = 0.0625$$

$$|P_5 - P_4| = |0.28125 - 0.3125| = 0.03125$$

$$|P_6 - P_5| = |0.265625 - 0.28125| = 0.015$$

$$|P_7 - P_6| = |0.2578125 - 0.265625| = 0.0078$$